Chapter 14 · Exponentials and logarithms

Starter 2

1. Evaluate when x = 3 and y = −1.

a. 5^x [a]

b. 3^y [b]

c. 2^{2x-1} [c]

d. 7^{1-y} [d]

e. 11^{x+3y} [e]

2. Write as a single power.

a. $6^8/6^2$ [a]

b. $y^3(y^9)^2$ [b]

c. $2^5 \times 2^9/2^8$ [c]

d. $\sqrt{x^8}$ [d]

3. Sketch a scatter graph and estimate a straight-line model.

x	1.2	2.1	3.5	4	5.8
y	5.8	7.4	9.4	10.3	12.8

[3: model]

[3: graph]

Chapter Overview 3

Exponential functions

Recognise exponential growth and decay, sketch graphs and differentiate exponential functions.

Logarithms

Use inverse functions, logarithm laws and natural logarithms to solve equations.

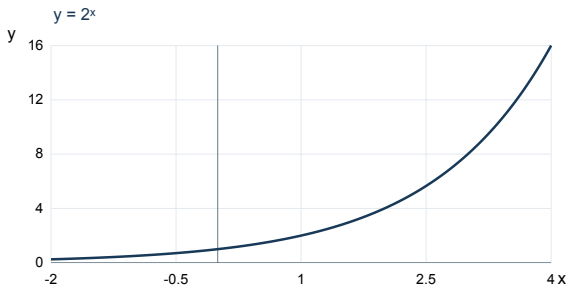
Modelling

Interpret model parameters and use logarithmic transformations to fit straight lines to nonlinear data.

y=2ˣ 4

$y = 2^x$

x	-2	-1	0	1	2	3	4
y	$1/4$	$1/2$	1	2	4	8	16

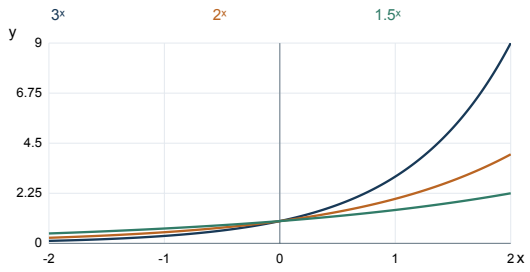


Each increase of 1 in x multiplies y by 2. The graph passes through (0,1), remains positive and approaches y = 0 as x decreases.

Contrasting exponential graphs 5

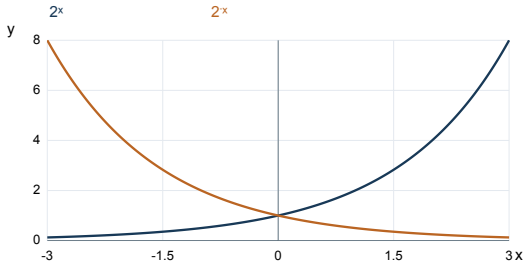
Compare $y = 3^x,\ y = 2^x,\ y = 1.5^x$.

All pass through (0,1). For positive x, the larger base grows faster; for negative x, the ordering reverses.



Compare $y = 2^x$ and $y = 2^{-x}$.

Relationship



Graph Transformations 6

Sketch $y = 2^{x+3}$. Describe its transformation from $y = 2^x$, and state its y-intercept.

Transformation

Intercept

Sketch

Exercise 14A (Pages 313-314) 7

6. The curve $y = ka^x$ passes through (1,6) and (4,48). Find k and a.

6

7. The curve $y = pq^x$ passes through $(-3,150)$ and (2,0.048). Find p and q and sketch the curve.

7

$y=e^x$ 8

The gradient of an exponential is proportional to its value.

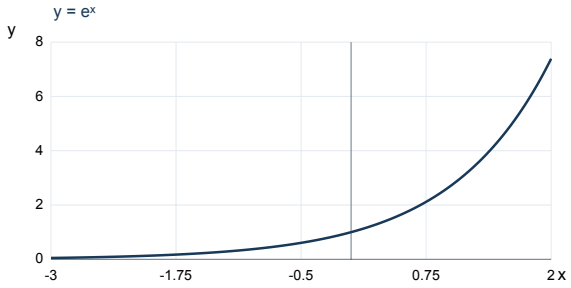
y	dy/dx (approximately)
1^x	0
1.5^x	$0.41(1.5^x)$
2^x	$0.69(2^x)$
2.5^x	$0.92(2.5^x)$
3^x	$1.10(3^x)$
3.5^x	$1.25(3.5^x)$

Is there a base for which the multiplier is exactly 1?

Special base

$e = 2.718281828\ldots$

The exponential function with base e has gradient equal to its own value.



$y = e^x \quad \Longrightarrow \quad \frac{dy}{dx} = e^x$

It passes through (0,1), where its gradient is 1, and has horizontal asymptote y = 0.

Differentiating $y=ae^{kx}$ 10

Differentiating an exponential

$\frac{d}{dx}(ae^{kx}) = ake^{kx}$

$y = e^{5x}$

1

$y = e^{-x}$

2

$y = 4e^{3x}$

3

More Graph Transformations 11

Sketch $y = e^{3x}$.

1

1: sketch

Sketch $y = 5e^{-x}$.

2

2: sketch

Sketch $y = 2 + e^{x/3}$.

3

3: sketch

Test Your Understanding 12

Sketch $y = e^{-2x} - 1$, stating its intercept and asymptote.

Sketch features

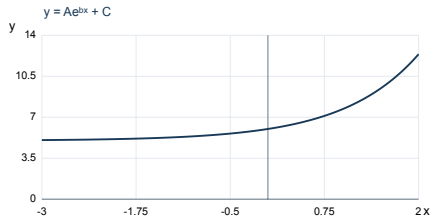
Sketch

Exercise 14B (Pages 316-317) 13

Each graph has the form $y = Ae^{bx} + C$. Find A and C, and state the sign of b.

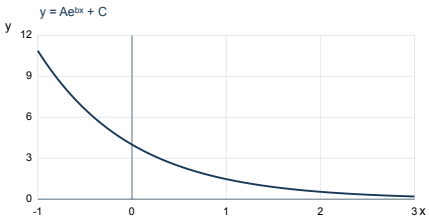
(a) Increasing; horizontal asymptote y = 5; y-intercept (0,6).

a



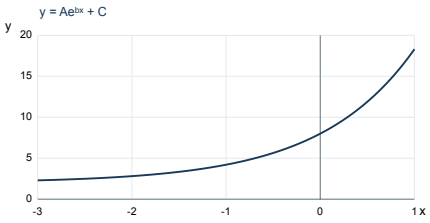
(b) Decreasing; horizontal asymptote y = 0; y-intercept (0,4).

b



(c) Increasing; horizontal asymptote y = 2; y-intercept (0,8).

c



Exercise 14B (Pages 316-317) 14

Differentiate each function.

1. e^{6x}

1

2. $e^{-x/3}$

2

3. $7e^{2x}$

3

4. $5e^{0.4x}$

4

5. $e^{3x} + 2e^{x}$

5

6. $e^{x}(e^{x} + 1)$

6

Exercise 14B (Pages 316-317) 15

Find the gradient of $y = e^{3x}$ at x = 2, 0 and -0.5.

Gradients

Find the tangent to $y = e^{0.2x}$ at (5,e).

Tangent

Just for your interest. . . 16

Where does e arise?

Invest £1 at a nominal annual interest rate of 100%, compounded in n equal instalments.

Balance after one year $= \left(1 + \frac{1}{n}\right)^n$

n	1	2	12	365	10000
Balance	2	2.25	2.6130	2.7146	2.7181

Limit

Exponential Modelling 17

A population is modelled by $P = 100e^{3t}$. Find its initial size and initial rate of growth.

Initial size

Initial rate

An exponential model multiplies by the same factor over equal time intervals. Its instantaneous rate of change is proportional to its current value.

Another Example 18

Pesticide density is $P = 160e^{-0.006t}$ mg/m², t days after application.

(a) Estimate P after 15 days.

a

(b) Interpret 160.

b

(c) Show that $dP/dt = kP$.

c

(d) Interpret the sign of k. (e) Sketch P against t.

d–e

Graph

Exercise 14C (Pages 318-319) 19

Air pressure at height h km is modelled by $p = e^{-0.13h}$ atmospheres.

(a) Estimate pressure at 4.394 km.

a

(b) Express dp/dh in terms of p.

b

(c) Explain how pressure changes with height.

c

(d) Find the percentage decrease for each additional kilometre.

d

Exercise 14C (Pages 318-319) 20

A £20,000 tractor depreciates according to two proposed models, t years after purchase.

$T_1 = 20000e^{-0.24t}$, $T_2 = 19000e^{-0.255t} + 1000$

(a) Compare their values after one year.

a

(b) Compare their values after ten years.

b

(c) Sketch both models. (d) Interpret the 1000 in model 2.

c–d

Graph

Logarithms undo exponentiation.

$a^x = n \iff \log_a n = x$

For real logarithms: $a > 0,\ a \neq 1,\ n > 0$.

Operation	Inverse operation
$3x$, $x+3$, $x^2\ (x \geq 0)$, x^5, 3^x	Inverses

$\log_a n = x$ (said "log base a of n") is equivalent to $a^x = n$.
The logarithm outputs the missing power. Here $a > 0$, $a \neq 1$ and $n > 0$.

$$3^2 = 9 \iff \log_3 9 = 2$$

Here are two methods of interchanging between these forms. Pick your favourite!

Method 1: Missing power

The base of the logarithm must match the base of the exponential.
$\log_2 8$ asks: "2 to what power gives 8?"

Imagine inserting the output of the logarithm just after its base. Click the button to move the missing power.

$$\log_2 8 = 3$$

Animation available on the online slide.

Method 2: Do the same operation to each side

Since KS3, you have used the same operation on each side of an equation to undo whatever you want to remove.
For example, subtract 2 from each side:
$3x + 2 = 11 \quad \Rightarrow \quad 3x = 9$.

"Log base a" and "a to the power of" undo each other: they are inverse functions.

$\log_2 8 = 3$
Apply $2^{\square}$ to each side:
$2^{\log_2 8} = 2^3 \quad \Rightarrow \quad 8 = 2^3$.

Examples 23

Evaluate each logarithm by finding the missing power.

1. $\log_5 25$

2. $\log_3 81$

3. $\log_4 1$

4. $\log_2 32$

5. $\log_4 4$

6. $\log_2(1/2)$

7. $\log_3(1/27)$

8. $\log_2(1/16)$

9. $\log_4(-1)$

10. $\log_{10} 1000$

11. $\log_a(a^3)$

With Your Calculator 24

Use a calculator. Give numerical answers to 4 significant figures.

1. $\log_3 7$

2. $\log_5 0.3$

3. $\ln 10$

4. $\ln e$

5. $\log_{10} 100$

6(a). Without a calculator, justify why $\log_2 50$ lies between 5 and 6. (b) Evaluate it to 4 significant figures.

6

7(a). Evaluate $\log_2 2,\ \log_3 3,\ \log_{17} 17$. (b) Explain the general rule.

7

8(a). Evaluate $\log_2 1,\ \log_3 1,\ \log_{17} 1$. (b) Explain the general rule.

8

Extension Exercise 26

1. [MAT 2015] Without a calculator, identify the largest number.

$$A = \frac{\sqrt{7}}{2},\ B = \frac{5}{4},\ C = \frac{\sqrt{10!}}{3(6!)}$$
$$D = \frac{\log_2 30}{\log_3 85},\ E = \frac{1+\sqrt{6}}{3}$$

1

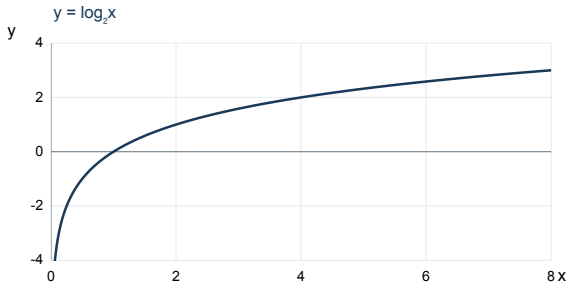
2. [MAT 2013] Positive a, b, c satisfy $\log_b a = 2$, $\log_b(c-3) = 3$, $\log_a(c+5) = 2$. Is a uniquely determined?

2

$y = \log_2 x$ 27

$y = \log_2 x$

x	$1/4$	$1/2$	1	2	4	8
y	-2	-1	0	1	2	3



The domain is $x > 0$; the range is all real numbers. The graph increases through $(1,0)$ and has vertical asymptote $x = 0$. It is the reflection of $y = 2^x$ in $y = x$.

Laws of Logs 28

Logarithm laws

For a valid base a and positive u, v:

$\log_a(uv) = \log_a u + \log_a v$

$\log_a(u/v) = \log_a u - \log_a v$

$\log_a(u^k) = k \log_a u$

Change of base

$$\log_a u = \frac{\log_b u}{\log_b a}$$

In particular, $\log_a 1 = 0$ and $\log_a a = 1$.

Examples

Combine into a single logarithm (1–4); expand using log laws (5–8). Assume positive variables and valid bases.

1. $\log_3 6 + \log_3 7$

1

2. $\log_2 15 - \log_2 3$

2

3. $2\log_5 3 + 3\log_5 2$

3

4. $\log_{10} 3 - 4\log_{10} 0.5$

4

5. $\log_a(x^2 y z^3)$

5

6. $\log_a(x/y^3)$

6

7. $\log_a(x\sqrt{y}/z)$

7

8. $\log_a(x/a^4)$

8

Anti Laws

These are NOT logarithm laws.

$\log_a(u+v) \neq \log_a u + \log_a v$ in general

Counterexample

$(\log_2 x)^3 \neq 3\log_2 x$ in general

Distinction

Solving Equations with Logs 31

Solve $\log_{10} 4 + 2\log_{10} x = 2$.

1

2

Test Your Understanding 32

Solve $2\log_2(x+15) - \log_2 x = 6$.

1

2

Solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

1

2

Given $a + b = 13$, $\log_6 a + \log_6 b = 2$ and a > b, find a and b.

Answer

Challenge: prove the quotient law for logarithms.

Proof

1. Solve $\log_3(x-7) - \frac{1}{2}\log_3 x = 1 - \log_3 2$.

1

2. Given $(\log_3 p)^2 = \log_3(p^2)$ and $\log_3(p+q) = \log_3 p + \log_3 q$, find p and q.

2

3. If $4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$, find the greatest a.

3

4. Prove $3/2 < \log_2 3 < 5/3$.

4

Solving equations with exponential terms 36

Solve using logarithms; give answers to 3 decimal places.

1. $3^x = 20$

1

2. $5^{4x-1} = 61$

2

Solving equations with exponential terms 37

Solve $3^x = 2^{x+1}$.

Take logarithms to any one consistent base.

1

2

Test Your Understanding 38

Solve each exponential equation.

1. $3^{2x-1} = 5$

2. $2^x 3^{x+1} = 5$

1

2

3. $3^{x+1} = 4^{x-1}$

3

Exercise 14F – Y1 Page 325

3(a). Solve $3^{x+1} = 2000$.

3a

4(a). Sketch $y = 4^x$. 4(b). Solve $4^{2x} - 10(4^x) + 16 = 0$.

4a

3(b). Solve $\log_5(x-3) = -1$.

3b

4b

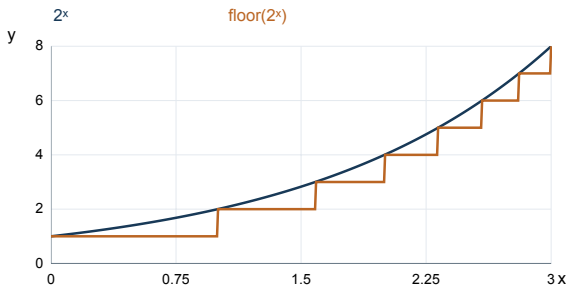
Extension [MAT 2011]: how many positive solutions satisfy $x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + 2$?

Extension

Graph

Very Challenging Extension!

[MAT 2013] Write $\lfloor x \rfloor$ for the greatest integer $\leq$ x. For a natural number n, evaluate $\int_0^n \lfloor 2^x \rfloor \, dx$.



On $\log_2 k \leq x < \log_2(k+1)$, the height is k. Put $N = 2^n$.

Rectangles

Simplify

Natural Logarithms 41

The natural logarithm is $\ln x = \log_e x$. It is the inverse of e^x.

$\ln(e^x) = x, \qquad e^{\ln x} = x \ (x > 0)$

Solve $e^x = 5$.

1

Solve $2\ln x + 1 = 5$.

2

Quadratics in e^x 42

Solve $e^{2x} + 2e^x - 15 = 0$.

1

Solve $e^x - 2e^{-x} = 1$.

2

A substitution $u = e^x$ turns each equation into a quadratic. Remember that u must be positive.

Test Your Understanding 43

Solve exactly.

1. $\ln(3x+1) = 2$

1

2. $e^{2x} + 5e^{x} = 6$

2

3. $2^{x}e^{x+1} = 3$

3

Exercise 14G – Y1 Page 327 44

4. Solve $e^{x} + 12e^{-x} = 7$ exactly.

4

6. Solve $3^{x}e^{4x-1} = 5$ exactly.

6

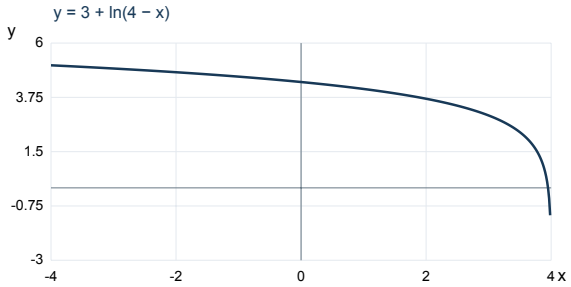
8. The graph is $y = 3 + \ln(4 - x)$. Find its exact intercepts A on the y-axis and B on the x-axis.

a

b

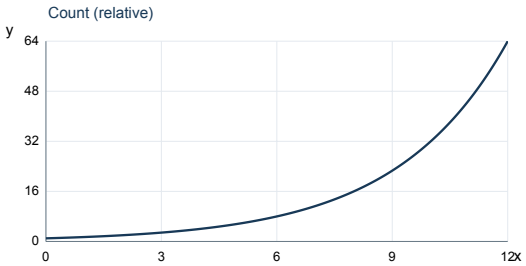
Challenge: $g(x) = Ae^{Bx} + C$ passes through (0,5) and (6,10), with asymptote y = 2. Find B.

Challenge



Exponential data can span a very wide range.

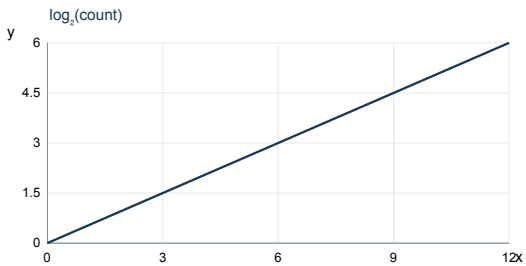
A historical computing model doubles transistor counts every two years. On an ordinary vertical scale, the smaller values cluster near the axis.



Taking logarithms turns multiplication into addition.

$\log_2(2y) = 1 + \log_2 y$

Every doubling adds 1 to the logged value. Plotting $\log_2$(count) against time gives a straight line under this model.



Graphs for Exponential Data 47

Logarithmic scales

A base-10 logarithmic scale compresses large ratios into manageable differences.

Earthquake amplitude

If the measured amplitude increases by a factor of 10, its base-10 logarithm increases by 1.

$\log_{10}(10A) = 1 + \log_{10} A$

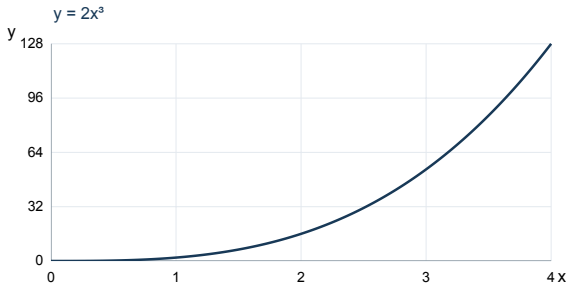
Do not confuse amplitude with energy.

The logarithmic amplitude comparison is not an energy comparison: these are different physical quantities.

Other Non-Linear Growth 48

Power-law growth: $y = 2x^3$, for x > 0.

A plot on ordinary axes curves upwards. Taking logarithms of BOTH variables produces a straight line.



$\log y = \log 2 + 3 \log x$

Plot log y vertically against log x horizontally: gradient 3 and intercept log 2.

Turning non-linear graphs into linear ones 49

Power-law model

$y = ax^n \Rightarrow \log y = n \log x + \log a$

Plot log y against log x. Gradient = n; intercept = log a. Require a > 0 and x > 0.

Exponential model

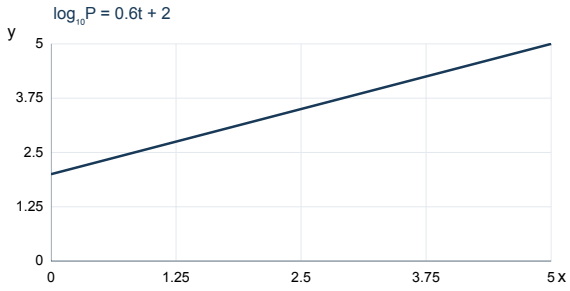
$y = ab^x \Rightarrow \log y = x \log b + \log a$

Plot log y against x. Gradient = log b; intercept = log a. Require a > 0 and b > 0.

Use one consistent logarithm base. Compare your transformed equation with Y = mX + c rather than memorising separate rules.

Example 50

A bacteria population has model $P = ab^t$. Its graph of $\log_{10}$P against t has gradient 0.6 and vertical intercept 2.



(a) Write the line equation.

a

(b) Find a and b.

b

(c) Interpret a.

c

Example 51

Historical city data are modelled by $P = aR^n$, where R is rank and P is population.

R	2	3	4	5	6
P	1000000	730000	620000	530000	480000

(a) Calculate $\log_{10}$R and $\log_{10}$P to 2 decimal places.

a

(b) Plot log P against log R and draw a best-fit line.
(c) Estimate a and n.

b–c

Graph

Test Your Understanding 52

A website models its followers by $P = ab^t$, t years after the start of 2015.

t	0.7	1.3	2.2
P	2353	3673	7162

(a) Calculate $\log_{10}$P to 3 decimal places.

a

(b) A best-fit line passes through the first and last transformed points. Its intercept is 3.147. Find its equation.

b

(c) Find the model parameters. (d) Predict P at t = 5.

c–d

Fitting model parameters to data is called regression. Logarithmic transformation lets us estimate a nonlinear model using a straight line.

5. Kleiber's law models metabolic rate by $R = am^b$. Mass m is in kg and rate R in kcal/day.

m	0.030	0.408	4.19	34.6	650
R	4.2	32.3	195	760	7637

(a) Calculate $\log_{10}$m and $\log_{10}$R to 2 decimal places.

a

(b) Plot log R against log m. (c) Estimate a and b from a best-fit line.

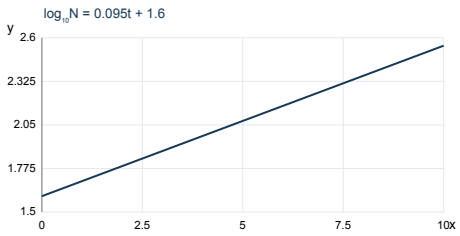
b–c

(d) Estimate the metabolic rate for mass 80 kg.

d

Graph

8. A virus model is $N = ab^t$. A graph of $\log_{10}$N against t passes through (0,1.6) and (10,2.55).



(a) Write the line equation.

a

(b) Find a and b to 2 significant figures.

b

(c) Interpret a.

c

(d) Predict the number after 30 days using the rounded model in (b).

d

(e) Why might the model overestimate later numbers?

e

Rounding matters: retaining the unrounded parameters gives about 28,000 at day 30. Always state which model is used.